The Cambrian Explosion: a theory

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Figure 1 Fig 1 Early Cambrian fauna (image licensed from Getty Images [UK]).

Introduction

It has long been suggested that the driver of the explosion in complex life during the Cambrian Period was the dramatic increase in atmospheric oxygen as a result of two 'Oxygen Events' during the Proterozoic Eon (Fig. 1).

This theory postulates that the substantial increase in oxygen levels during this time resulted in an extinction event that killed off most anaerobic life in Earth's oceans and enabled the evolution of complex oxygen-breathing organisms to fill the resultant faunal gap.

Such extinction events have occurred several times during Earth's history, and in each case the empty ecological niches were filled by new life. For example, the end-Permian extinction that killed off all large amphibians paved the way for the rise of reptiles; and the end-Cretaceous extinction that killed off the dinosaurs and marine reptiles paved the way for the rise of mammals.

However, was oxygen the only driver at the time of the 'Cambrian Explosion'? There is a second possibility that could have made not only a major contribution, but also provides an explanation of why complex life failed to develop earlier in Earth's history: the absence or presence of an ozone layer in Earth's atmosphere.

Atmosphere of the early Earth

During the Archean Eon (4 Ga–2.5 Ga) Earth's atmosphere contained only carbon dioxide (CO_2), methane (CH_4), water vapour (H_2O), carbon monoxide (CO), a little nitrogen (N) and hydrogen (H) (Borzenkova and Turchinovich 2009).

Fauna of the early Earth

During the Archean and much of the Proterozoic most lifeforms were of marine purple and green sulphur bacteria and algae, blue-green stromatolites and multicellular metazoans, all of which utilised anaerobic photosynthesis to generate energy, i.e. sunlight used to split molecules of water, sulphur, iron or hydrogen sulphide (Nalin 2013).

Unlike oxygenic photosynthesis, the waste products produced through anaerobic photosynthesis are carbon dioxide and methane instead of oxygen.

Oxygenation events

The Great Oxygenation Event (GOE) occurred during the Palaeoproterozoic Era *c.* 2.4 Ga– 2.0 Ga. The result of this event was to increase free oxygen in Earth's atmosphere from 0.0001% to *c.* 1–2% (Ligrone 2019).

The Neo-Proterozoic (also called the 'Second') Oxygenation Event (NOE) occurred during the Neoproterozoic Era *c.* 0.85 Ga–0.54 Ga. The result of this event was to increase free oxygen in Earth's atmosphere from 1–2% to *c.* 10% (Holland 2006) (Fig. 2).

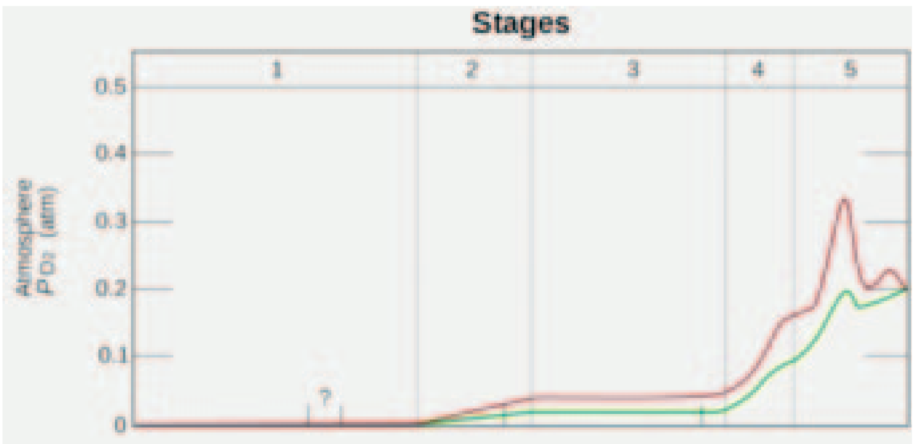
Importance of ozone

Solar UV radiation covers the wavelength range 100–400 nm and is divided into three bands:

UV-A (315–400 nm): the most damaging type of UV at the Earth's surface
UV-B (280–315 nm): cannot penetrate beyond superficial skin layers
UV-C (100–280 nm): potentially the most damaging type of ultra-violet radiation of all (World Health Organisation 2016).

Currently 50% of UV-A, 90% of UV-B-and 100% of UV-C is absorbed by the ozone layer (National Aeronautics and Space Administration 2021) (Fig. 3, *overleaf*). Studies by the US National Aeronautics and Space Administration (NASA) show

Figure 2 Estimated evolution of atmospheric PO2. The upper red and lower green lines represent the range of the estimates. The stages are: stage 1 (3.85 Ga–2.45 Ga); stage 2 (2.45 Ga–1.85 Ga); stage 3 (1.85 Ga–0.85 Ga); stage 4 (0.85 Ga–0.54 Ga); and stage 5 (0.54 Ga–present) (© Heinrich D. Holland, reproduced with permission under GNU: Free Documentation License).



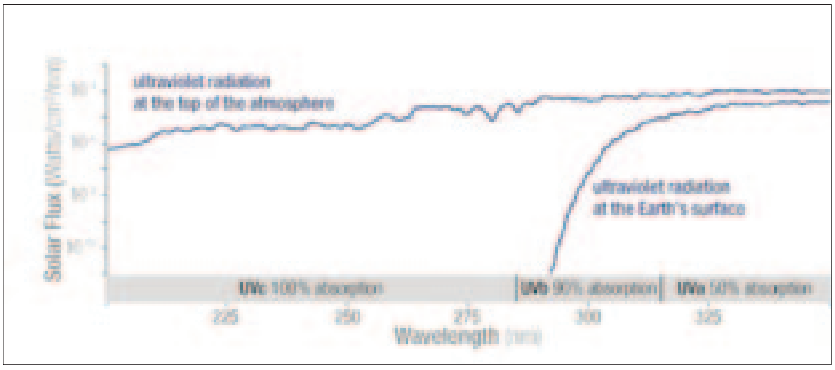


Figure 3 Solar UV levels and absorbency by the ozone layer (permission courtesy of NASA Ozone Watch).

that today the ozone represents only 0.00006% of the atmosphere, while oxygen is *c.* 20%. The results of these studies suggest that before the NOE a 1–2% oxygen saturation would have meant that ozone was only produced at levels of *c.* 0.000005%; that is, 92.5% less than the level today.

Even taking into account the fact that during the time interval of 4.5 Ga to 1 Ga, solar luminosity was 5% to 20% lower than at present, with no ozone layer, levels of solar UV radiation reaching the surface of the Earth before the NOE was likely to have been up to 1,000,000 times greater than it is today (National Aeronautics and Space Administration 2010; Blumthaler, Ambach and Ellinger 1997; World Health Organisation 2002).

Effect of solar UV radiation on biology

It is estimated that during the Proterozoic, marine organisms were most abundant in coastal and ocean environments within the first 5 m to 25 m depth zone (Butterfield 2007; Avila-Alonso, Baetens, Cardenas and De Baets 2017).

Today UV radiation penetrates seawater to a depth of between 20 m and 40 m, depending on the clarity of the water. But with no ozone layer to absorb UV radiation, penetration is likely to have been to much deeper depths (Fleischmann 1989).

The effects of UV radiation on living organisms include the destruction or inactivation of proteins, the destruction of cell membranes, and damage to strands of ribonucleic acid (RNA) and deoxyribonucleic acid (DNA), causing mutations and cell death (Sutherland 2006; Gao *et al.* 2019; Sinha and Hader 2002) (Fig 4).

UV-A and UV-B radiation also inhibits photosynthetic production, and therefore also reduces carbon dioxide intake. Studies have shown that UV-A, in particular, can significantly decrease photosynthesis by more than 70% (White and Jahnke 2002).

Conclusion

With no free oxygen in the Earth's atmosphere up to the end of the Archean Eon the triatomic allotrope of oxygen produced by the reaction of solar ultra-violet radiation on oxygen molecules — ozone — could not have existed (Chapman 1930).

Following the Great Oxygenation Event, the amount of ozone that could have been produced from the very limited quantities of free oxygen available in the atmosphere provided only a marginal reduction in the levels of UV radiation reaching the Earth's surface.

Continual exposure to extremely high levels of solar UV radiation — especially UV-C radiation that today no longer reaches the Earth's surface — is likely to have inhibited critical life functions and to have produced high levels of genetic damage. Both these factors would have had a significant detrimental effect on the health, longevity, reproductive capability and evolution of Precambrian organisms.

Only after the Neo-Proterozoic Oxygenation Event *c.* 0.85 Ga–0.54 Ga could an effective ozone layer have been formed and created an environment that enabled Cambrian organisms to develop and flourish.

A final thought: Can it be just a coincidence that eyes — organs particularly susceptible to serious damage from UV-C radiation — evolved from simple photoreceptors only after the Neo-Proterozoic Oxygenation Event? The evidence suggests perhaps not (Hazlehurst and Hendry 2017).

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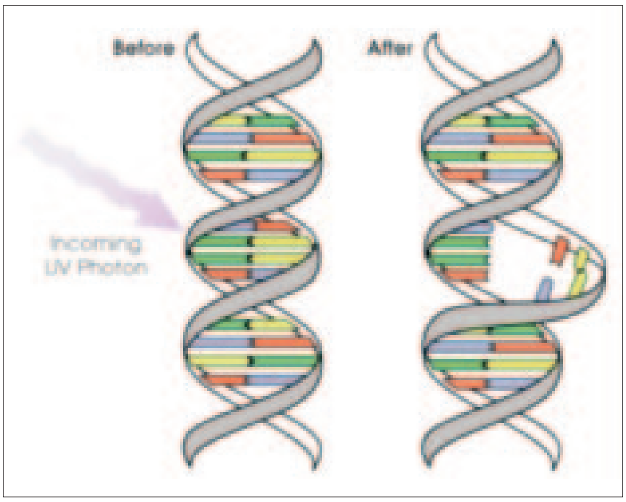
Figure 4 UV radiation damage to DNA (© David Herring: I, the copyright holder of this work, release this work into the public domain. This applies worldwide).



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